

1- Consider following 3-bit image:

f =

3 4 5

2 2 6

Apply a mean filter to the image, considering a filter window size of 3x3.

Solution:

Padding image:

0 0 0 0 0

0 3 4 5 0

0 2 2 6 0

0 0 0 0 0

$$\hat{f}(0,0) = \frac{0+0+0+0+3+4+0+2+2}{9} = 1.2 \quad , \quad \hat{f}(0,0) = \text{round}(1.2) = 1$$

$$\hat{f}(0,1) = (0 + 0 + 0 + 3 + 4 + 5 + 2 + 2 + 6)/9 = 2.44 \quad , \quad \hat{f}(0,1) = \text{round}(2.44) = 2$$

$\hat{f}$ =

1 2 2

1 2 2

2-Apply geometric mean filter and find the element $\hat{f}(1,1)$ for the flowing image. Use filter window size of 3x3.

f =

3 4 5

2 2 6

5 3 7

Solution:

$$\hat{f}(1,1) = (3 * 4 * 5 * 2 * 2 * 6 * 5 * 3 * 7)^{1/(3*3)} = (151200)^{1/9} = 3.7628 \quad , \quad \hat{f}(1,1) = \text{round}(3.76) = 4$$

3- Apply Harmonic mean filter to estimate $\hat{f}(1,1)$ for image in question 2. Use filter window size of 3x3.

$$\hat{f}(1,1) = \frac{mn}{\sum_{(s,t) \in S_{xyg(s,t)}} \frac{1}{f(s,t)}} \quad , \quad \hat{f}(1,1) = \frac{3(3)}{\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{2} + \frac{1}{2} + \frac{1}{6} + \frac{1}{5} + \frac{1}{3} + \frac{1}{7}} = \frac{9}{2.6262} = 3.4270, \quad \hat{f}(1,1) = \text{round}(3.427) = 3$$

4- Consider flowing image, Use filter window size of 3x3.

f =

3	4	5
2	2	6
5	3	7

- A. Apply Median filter and calculate the element $\hat{f}(1,1)$.
- B. Apply Max filter and calculate the element $\hat{f}(1,1)$.
- C. Apply Min filter and calculate the element $\hat{f}(1,1)$.
- D. Apply Midpoint filter and calculate the element $\hat{f}(1,1)$.

Solution:

A: $\hat{f}(x, y) = \text{median}_{(s,t) \in S_{xy}} \{g(s, t)\}$

$\text{Sort}(g(s, t)) = 2, 2, 3, 3, 4, 5, 5, 6, 7,$ $\hat{f}(1,1) = 4$

B: $\hat{f}(x, y) = \max_{(s,t) \in S_{xy}} \{g(s, t)\}$, $\hat{f}(1,1) = 7$

C: $\hat{f}(x, y) = \min_{(s,t) \in S_{xy}} \{g(s, t)\}$, $\hat{f}(1,1) = 2$

D: $\hat{f}(x, y) = \frac{1}{2} \left\{ \max_{(s,t) \in S_{xy}} \{g(s, t)\} + \min_{(s,t) \in S_{xy}} \{g(s, t)\} \right\}$, $\hat{f}(1,1) = \frac{7+2}{2} = 4.5$, $\hat{f}(1,1) = \text{round}(4.5) = 5$

5- An 8-bit image is given as

g=

245	239	238	328
234	235	228	213
225	235	232	239

Standard deviation of any pixel noise is $\sigma_\eta = 3$, the size of filter's window is 3×3 . Use adaptive filter algorithm to estimate the gray level of pixel (1,1).

$$\hat{f}(x, y) = g(x, y) - \frac{\sigma_\eta^2}{\sigma_L^2} (g(x, y) - m_L)$$

$$m_L(1,1) = \frac{245 + 239 + 238 + 234 + 235 + 228 + 225 + 235 + 232}{9} = 234.5556$$

$$\begin{aligned} \sigma_L^2 = \frac{1}{9} \{ & (245 - 234.5556)^2 + (239 - 234.5556)^2 + (238 - 234.5556)^2 + (234 - 234.5556)^2 \\ & + (235 - 234.5556)^2 + (228 - 234.5556)^2 + (225 - 234.5556)^2 + (235 - 234.5556)^2 \\ & + (232 - 234.5556)^2 \} = 31.3580 \end{aligned}$$

$$\hat{f}(1,1) = 235 - \frac{3^2}{31.3580} (235 - 234.5556) = 234.87, \quad \hat{f}(1,1) = \text{round}(234.87) = 235$$

- 6- A 3-bit image with a size of 120×100 pixels is available. The number of pixels for each intensity level is given as follows:

Intensity level	Number of pixels
0	1000
1	1500
2	2500
3	2700
4	1600
5	1200
6	1000
7	500
total	12000

Solution:

Intensity level $z_i, i = 0, 1, \dots, 7$	Number of pixels	Probability $p(z_i)$	meanType equation here.
$z_0 = 0$	1000	0.0833	
$z_1 = 1$	1500	0.1250	
$z_2 = 2$	2500	0.2083	
$z_3 = 3$	2700	0.2250	
$z_4 = 4$	1600	0.1333	
$z_5 = 5$	1200	0.1000	
$z_6 = 6$	1000	0.0833	
$z_7 = 7$	500	0.0417	
total	12000	1	

$$\text{mean} = \bar{z} = \sum_{i=0}^7 z_i p(z_i) = (0(0.0833) + 1(0.1250) + \dots + 7(0.0417)) = 3.0417$$

$$\text{Variance} = \sigma^2 = \sum_{i=0}^7 (z_i - \bar{z})^2 p(z_i) = ((0 - 3.0417)^2(0.0833) + (1 - 3.0417)^2(0.1250) + \dots + (7 - 3.0417)^2(0.0833)) = 3.4066$$

$$\text{Standard deviation} = \sigma = \sqrt{3.4066} = 1.8457$$